

Unit 3 - Data Estimation-Exercises

1. Mean and Population Variance

Step 1: Write the data

Delivery times:

20, 25, 30, 35, 40

Number of observations:

$$N = 5$$

Step 2: Calculate the mean

Formula:

$$\bar{x} = \Sigma x / N$$

$$\bar{x} = (20 + 25 + 30 + 35 + 40) / 5$$

$$\bar{x} = 150 / 5$$

$$\bar{x} = 30$$

Step 3: Calculate deviations from the mean

x	$x - \bar{x}$	$(x - \bar{x})^2$
20	-10	100
25	-5	25
30	0	0
35	5	25
40	10	100
Total		250

Step 4: Calculate population variance

Formula:

$$\sigma^2 = \Sigma(x - \bar{x})^2 / N$$

$$\sigma^2 = 250 / 5$$

$$\sigma^2 = 50$$

Final Answer

Mean = 30 minutes

Population Variance = 50 minutes²

2. Why Sampling is Used

Step 1: Identify the population

Total university students:

$N = 5000$

Step 2: Identify the sample

Random sample:

$n = 100$

Step 3: Why use sampling?

Surveying all 5,000 students would require:

- More time
- More money
- More effort
- More data collection and processing

A random sample of 100 students is much easier to manage.

Step 4: Why is a random sample useful?

If the 100 students are selected randomly, they can provide a reasonable representation of the entire student population.

The average study time calculated from the sample can then be used to estimate the average study time of all 5,000 students.

Final Answer

Sampling is used because studying all 5,000 students would be time-consuming, expensive, and difficult to manage. A properly selected random sample of 100 students is more efficient and can provide a reasonable estimate of the population average.

3. Sample Proportion

Step 1: Write the given values

Total customers:

$$n = 200$$

Satisfied customers:

$$x = 150$$

Step 2: Write the sample proportion formula

$$\hat{p} = x / n$$

Step 3: Substitute the values

$$\hat{p} = 150 / 200$$

$$\hat{p} = 0.75$$

Step 4: Convert into percentage

$$0.75 \times 100 = 75\%$$

Final Answer

Sample Proportion = 0.75

Percentage of satisfied customers = 75%

4. Covariance Between X and Y

Step 1: Calculate the means

Mean of X:

$$\bar{X} = (2 + 4 + 6 + 8) / 4$$

$$\bar{X} = 20 / 4$$

$$\bar{X} = 5$$

Mean of Y:

$$\bar{Y} = (5 + 7 + 9 + 11) / 4$$

$$\bar{Y} = 32 / 4$$

$$\bar{Y} = 8$$

Step 2: Calculate deviations and products

X	Y	$X - \bar{X}$	$Y - \bar{Y}$	$(X - \bar{X})(Y - \bar{Y})$
2	5	-3	-3	9
4	7	-1	-1	1
6	9	1	1	1
8	11	3	3	9
				20

Step 3: Calculate covariance

Using the population covariance formula:

$$\text{Cov}(X, Y) = \Sigma(X - \bar{X})(Y - \bar{Y}) / N$$

$$\text{Cov}(X, Y) = 20 / 4$$

$$\text{Cov}(X, Y) = 5$$

Step 4: Determine the relationship

The covariance is:

$$5 > 0$$

Since covariance is positive, X and Y move in the same direction.

As X increases, Y also increases.

Final Answer

$$\text{Cov}(X, Y) = 5$$

The relationship is positive.

5. Covariance Between Exercise Hours and Calories Burned

Step 1: Calculate the means

Mean of X:

$$\bar{X} = (1 + 2 + 3 + 4 + 5) / 5$$

$$\bar{X} = 15 / 5$$

$$\bar{X} = 3$$

Mean of Y:

$$\bar{Y} = (200 + 300 + 400 + 500 + 600) / 5$$

$$\bar{Y} = 2000 / 5$$

$$\bar{Y} = 400$$

Step 2: Calculate deviations and products

X	Y	$X - \bar{X}$	$Y - \bar{Y}$	Product
1	200	-2	-200	400
2	300	-1	-100	100
3	400	0	0	0
4	500	1	100	100
5	600	2	200	400
				1000

Step 3: Calculate covariance

Population covariance formula:

$$\text{Cov}(X, Y) = \Sigma(X - \bar{X})(Y - \bar{Y}) / N$$

$$\text{Cov}(X, Y) = 1000 / 5$$

$$\text{Cov}(X, Y) = 200$$

Step 4: Interpret the result

Since:

$$200 > 0$$

the covariance is positive.

Therefore, as exercise hours increase, calories burned also increase.

Final Answer

$$\text{Cov}(X, Y) = 200$$

The relationship is positive: more exercise hours are associated with more calories burned.

6. Pearson's Correlation Coefficient

Step 1: Calculate the means

Mean of X:

$$\bar{X} = (1 + 2 + 3 + 4 + 5) / 5$$

$$\bar{X} = 15 / 5$$

$$\bar{X} = 3$$

Mean of Y:

$$\bar{Y} = (45 + 50 + 55 + 65 + 70) / 5$$

$$\bar{Y} = 285 / 5$$

$$\bar{Y} = 57$$

Step 2: Calculate deviations

X	Y	$X - \bar{X}$	$Y - \bar{Y}$
1	45	-2	-12
2	50	-1	-7
3	55	0	-2
4	65	1	8
5	70	2	13

Step 3: Calculate required products

$X - \bar{X}$	$Y - \bar{Y}$	Product	$(X - \bar{X})^2$	$(Y - \bar{Y})^2$
-2	-12	24	4	144
-1	-7	7	1	49
0	-2	0	0	4
1	8	8	1	64
2	13	26	4	169
Total		65	10	430

Therefore:

$$\Sigma(X - \bar{X})(Y - \bar{Y}) = 65$$

$$\Sigma(X - \bar{X})^2 = 10$$

$$\Sigma(Y - \bar{Y})^2 = 430$$

Step 4: Write Pearson's correlation formula

$$r = \Sigma(X - \bar{X})(Y - \bar{Y}) / \sqrt{[\Sigma(X - \bar{X})^2 \Sigma(Y - \bar{Y})^2]}$$

Step 5: Substitute the values

$$r = 65 / \sqrt{(10 \times 430)}$$

$$r = 65 / \sqrt{4300}$$

$$\sqrt{4300} \approx 65.574$$

$$r = 65 / 65.574$$

$$r \approx 0.991$$

Step 6: Interpret the result

Since $r \approx 0.991$ and r is very close to $+1$, there is a very strong positive correlation.

This means that as study hours increase, marks tend to increase strongly.

Final Answer

$r \approx 0.991$

The dataset has a very strong positive relationship between study hours and marks.